

DIABETES READMISSION CLINICAL AI
CLINICAL AI MODEL VALIDATION REPORT

Project: Diabetes Readmission Clinical AI
Project Descriptor: 30-Day Hospital Readmission Clinical Decision-Support Prototype
Validation Stage: Internal Locked-Test Validation
Model ID: DIABETES_READMISSION_XGB_D13_V1
Model Version: 1.0.0
Frozen Operating Threshold: 0.12
Evaluation Mode: One-Time Confirmatory Evaluation
Document Status: Final
Production Clinical Use: NOT AUTHORIZED

1. EXECUTIVE VALIDATION SUMMARY

The **Diabetes Readmission Clinical AI** is an end-to-end clinical machine-learning project designed to estimate the probability of hospital readmission within 30 days among eligible adult diabetes-related hospital encounters and support clinician prioritization for additional readmission-prevention review.

This report documents the model's **confirmatory internal locked-test validation** following completion of model development, preprocessing freeze, operating-threshold selection, robustness assessment, subgroup assessment, explainability analysis, and model registration.

The registered XGBoost model, preprocessing pipeline, feature contract, and **0.12 operating threshold were frozen before locked-test evaluation**. No retraining, threshold optimization, feature redesign, or post hoc model modification was performed using locked-test outcomes.

The locked-test cohort contained **15,038 encounters**, including **1,707 30-day readmissions**, representing an outcome prevalence of **11.35%**.

Key Validation Results

Measure	Locked-Test Result
ROC-AUC	0.623
PR-AUC	0.182
Brier Score	0.0983
Sensitivity	48.74%
Specificity	70.54%
Positive Predictive Value	17.48%
Negative Predictive Value	91.49%
Alert Rate	31.65%
Alerts per 100 Encounters	31.65
Number Needed to Evaluate	5.72

Aggregate performance remained broadly stable relative to the preceding validation partition. However, validation also confirmed important limitations, including **limited discrimination, material dependence on prior healthcare-utilization information, subgroup and admission-context heterogeneity, unresolved external transportability, and absence of established clinical effectiveness**.

Validation Decision

CONDITIONAL PASS — PROGRESS WITH DOCUMENTED RESIDUAL RISKS

The model may progress to research-stage clinical translation, deployment architecture, and monitoring design.

This decision **does not authorize production clinical deployment**.

Evidence Domain	Status
Internal Locked-Test Validation	COMPLETED
Robustness Assessment	COMPLETED — RESIDUAL RISKS IDENTIFIED
Explainability Assessment	COMPLETED — RESIDUAL RISKS IDENTIFIED
Subgroup/Fairness Assessment	COMPLETED — HETEROGENEITY DOCUMENTED
External Validation	NOT ESTABLISHED
Clinical Effectiveness	NOT ESTABLISHED
Production Clinical Deployment	NOT AUTHORIZED

2. MODEL, INTENDED USE AND VALIDATION DESIGN

2.1 Model Configuration

The evaluated model was frozen before confirmatory evaluation.

Component	Validated Configuration
Registry ID	DIABETES READMISSION XGB D13 V1
Algorithm	XGBoost Classifier
Model Version	1.0.0
Preprocessing	Frozen D7 preprocessing pipeline
Source Features	10
Transformed Features	49
Operating Threshold	0.12
Validation Partition	Locked Test
Evaluation Mode	One-Time Confirmatory Evaluation

The model consumes ten governed source features covering patient characteristics, admission context, and prior healthcare utilization: **race, gender, age, admission type, admission source, prior outpatient use, prior emergency use, prior inpatient use, prior utilization intensity, and prior utilization domain count**.

The frozen preprocessing pipeline transforms these into **49 model-ready features**.

2.2 Intended Use

The model is intended as a **clinician-facing prioritization decision-support prototype** for identifying eligible patients who may warrant additional human review for 30-day readmission-prevention interventions.

It provides:

- an estimated probability of 30-day readmission; and
- an advisory prioritization signal based on the frozen threshold.

The model does **not** independently determine diagnosis, treatment, discharge, eligibility for care, restriction of care, or other autonomous clinical decisions.

Clinical decision authority remains with qualified healthcare professionals.

2.3 Validation Design

The evaluation used a **locked-test confirmatory design**.

The test partition was withheld from model development and candidate selection and evaluated only after model configuration and operating threshold were frozen.

The evaluation therefore provides strong internal confirmatory evidence but remains **internal validation**, because the test population originates from the same underlying source environment as model development.

Locked-test validation is not external validation.

3. VALIDATION POPULATION

The project originated from the UCI diabetes hospital readmission dataset containing **101,766 hospital encounters and 50 source variables**.

Following governed cohort construction, the analytical population contained:

- **69,944 eligible encounters;**
- **49,307 unique patients;**
- **7,907 30-day readmissions;** and
- an overall readmission prevalence of approximately **11.30%**.

Locked-Test Cohort

Measure	Result
Encounters	15,038
30-Day Readmissions	1,707
Non-Readmissions	13,331
Outcome Prevalence	11.35%

Locked-test prevalence was closely aligned with the preceding validation prevalence of approximately **11.24%**, supporting population-level stability between the two internal evaluation partitions.

4. MODEL PERFORMANCE

4.1 Discrimination and Probability Performance

Metric	Result
ROC-AUC	0.6234
PR-AUC	0.1820
Brier Score	0.0983
Log Loss	0.3433

The model demonstrated **limited-to-moderate discrimination**.

Given the relatively low outcome prevalence, PR-AUC was evaluated alongside ROC-AUC to provide a more informative assessment of performance for the minority readmission outcome.

The observed discrimination supports use only within the model's current **human-supervised research and validation context** and does not support autonomous clinical decision-making.

4.2 Frozen Operating-Point Performance

At the frozen threshold of **0.12**:

Readmission	No Readmission
TP = 832	FP = 3,927
FN = 875	TN = 9,404

Priority Flag

Metric	Result
Sensitivity	48.74%
Specificity	70.54%
PPV	17.48%
NPV	91.49%
F1 Score	25.73%

The model identified approximately **49% of observed readmissions**, while approximately **29% of encounters without readmission** generated a priority signal.

The NPV shall not be interpreted as evidence that a non-prioritized patient is clinically low risk, particularly given the identified utilization-dependent failure mode.

4.3 Operational Workload

The frozen threshold generated **4,759 priority alerts**.

Measure	Result
Alert Rate	31.65%
Alerts per 100 Encounters	31.65
Number Needed to Evaluate	5.72

Approximately **32 of every 100 eligible encounters** would therefore be prioritized for additional review.

These measures quantify potential workflow burden; they do not establish clinical effectiveness, operational acceptability, or cost-effectiveness.

5. VALIDATION STABILITY

Locked-test performance was compared with the preceding validation partition to assess whether model behavior materially deteriorated when evaluated against the previously unseen confirmatory partition.

Metric	Validation	Locked Test	Difference
Prevalence	11.24%	11.35%	+0.11 pp
PR-AUC	0.1849	0.1820	-0.0028
ROC-AUC	0.6204	0.6234	+0.0030
Brier Score	0.0974	0.0983	+0.0009
Sensitivity	46.39%	48.74%	+2.35 pp
Specificity	70.63%	70.54%	-0.09 pp
PPV	16.67%	17.48%	+0.81 pp
NPV	91.23%	91.49%	+0.26 pp
Alert Rate	31.28%	31.65%	+0.36 pp
NNE	6.00	5.72	-0.28

No material aggregate performance collapse was observed.

This supports **technical stability within the internal source environment**, but does not establish stability across different hospitals, populations, geographies, healthcare systems, or future clinical workflows.

6. ROBUSTNESS, EXPLAINABILITY AND SAFETY FINDINGS

6.1 Utilization-Dependent Model Behavior

Robustness testing identified material dependence on prior healthcare-utilization information.

Of particular importance, patients with:

- zero recorded prior outpatient utilization;
- zero recorded prior emergency utilization; and
- zero recorded prior inpatient utilization

demonstrated **0% sensitivity within the evaluated locked-test utilization profile at the frozen threshold**.

This represents a material safety limitation.

The clinical translation prototype therefore implements an explicit warning that absence of a priority flag in patients without recorded prior utilization **must not be interpreted as absence of readmission risk**.

Residual Risk: UTILIZATION_DEPENDENT_MODEL_BEHAVIOR — MATERIAL

6.2 Explainability

SHAP analysis identified substantial model reliance on prior-utilization-related feature families.

Evaluation Stage	Utilization-Related Attribution
D12 Validation	68.52%
D14 Locked Test	77.27%

The increase in attribution concentration during locked-test evaluation reinforces the utilization-dependence finding.

SHAP outputs are interpreted as **model contribution, not clinical causality**.

Residual Risk: EXPLAINABILITY_UTILIZATION_DOMINANCE — MATERIAL

6.3 Subgroup and Admission-Context Performance

Model operating characteristics varied across relevant patient subgroups and admission contexts.

The evidence did not establish uniform performance across all evaluated populations.

These findings are retained as:

- SUBGROUP_OPERATING_HETEROGENEITY — DOCUMENTED
- ADMISSION_CONTEXT_HETEROGENEITY — DOCUMENTED

Rather than concealing these differences behind aggregate performance, they are carried forward into monitoring and future validation requirements.

6.4 Clinical Safety

Key safety considerations include:

- false-negative predictions;
- false-positive alert burden;
- inappropriate reassurance from absence of an alert;
- automation bias;
- utilization-dependent behavior;
- subgroup/context heterogeneity;
- use outside the validated population; and
- interpretation of model attribution as clinical causality.

These findings reinforce the requirement for **human oversight and restricted intended use**.

7. RESIDUAL RISK AND EVIDENCE GAPS

Seven residual risks remain following confirmatory validation.

Residual Risk	Status
Utilization-Dependent Model Behavior	MATERIAL
Explainability Utilization Dominance	MATERIAL
Subgroup Operating Heterogeneity	DOCUMENTED
Admission-Context Heterogeneity	DOCUMENTED

Limited Model Discrimination	MATERIAL
External Transportability	UNRESOLVED
Clinical Effectiveness	UNRESOLVED

External Validation

External validation is **not established**.

Because the locked-test population originates from the same underlying source environment as development, the current evidence does not establish geographic, institutional, health-system, or population transportability.

A future external validation should evaluate the **frozen model and threshold before external-data-driven retraining or recalibration**, where scientifically feasible.

Clinical Effectiveness

Clinical effectiveness is **not established**.

The present validation does not demonstrate that use of the model reduces readmissions, improves patient outcomes, improves discharge planning, reduces cost, or produces net clinical benefit.

Such claims require additional prospective and clinical-effectiveness evidence.

8. GOVERNANCE DECISION

The locked-test evaluation demonstrated that:

- the registered frozen model executed successfully against the confirmatory partition;
- aggregate performance remained broadly stable relative to preceding validation;
- frozen-threshold operating behavior remained stable;
- no major internal performance collapse was identified;
- material utilization dependence remains;
- subgroup and admission-context heterogeneity remain documented;
- discrimination remains limited;
- external transportability remains unresolved; and
- clinical effectiveness remains unestablished.

Validation Gate

Gate Integrity: PASS

Evaluation Mode: ONE_TIME_CONFIRMATORY_EVALUATION

Partition: LOCKED_TEST

Governance Disposition

CONDITIONAL PASS — PROGRESS TO DEPLOYMENT AND MONITORING DESIGN WITH DOCUMENTED RESIDUAL RISKS

Progression is conditional upon:

- continued human oversight;
- preservation of the frozen model and threshold unless formal change control is initiated;

- retention of identified runtime safety warnings;
- residual-risk traceability;
- monitoring of utilization dependence and subgroup/context heterogeneity;
- external validation before transportability claims;
- additional evidence before clinical-effectiveness claims; and
- separate governance authorization before production clinical deployment.

Current Evidence Position

Domain	Final Status
Internal Locked-Test Validation	COMPLETED
Model Stability	SUPPORTED WITHIN INTERNAL SOURCE ENVIRONMENT
Robustness	COMPLETED WITH MATERIAL LIMITATIONS
Explainability	COMPLETED WITH MATERIAL UTILIZATION DEPENDENCE
Subgroup/Fairness	HETEROGENEITY DOCUMENTED
External Validation	NOT ESTABLISHED
Prospective Clinical Validation	NOT ESTABLISHED
Clinical Effectiveness	NOT ESTABLISHED
Production	CLINICAL
Deployment	NOT AUTHORIZED

9. VALIDATION CONCLUSION AND NEXT EVIDENCE GATE

The **Diabetes Readmission Clinical AI** successfully completed its prespecified internal locked-test confirmatory evaluation using the registered XGBoost model, frozen preprocessing pipeline, and frozen operating threshold of **0.12**.

Performance remained broadly stable relative to preceding validation, providing sufficient evidence for continued **research-stage clinical translation, governance, deployment architecture, and monitoring design**.

The evaluation also demonstrates why clinical AI validation cannot stop at a single performance metric. Limited discrimination, utilization-dependent behavior, explainability concentration, subgroup heterogeneity, operational alert burden, and unresolved transportability were retained as explicit evidence limitations rather than obscured.

Next Evidence Priorities:

1. External Validation

Evaluate the frozen model in an appropriately independent institution, population, healthcare system, geography, or data source.

2. External Calibration and Transportability

Assess whether predicted probabilities and frozen-threshold performance remain stable outside the development environment.

3. External Subgroup and Robustness Validation

Reassess utilization dependence, demographic performance, and admission-context heterogeneity.

4. Prospective Clinical Evaluation

Evaluate model behavior, clinician interaction, workflow integration, overrides, safety events, and operational impact prospectively.

5. Clinical Effectiveness

Determine whether model-supported intervention produces measurable patient, clinical, workflow, or economic benefit.

Final Decision

CONDITIONAL PASS FOR CONTINUED RESEARCH, CLINICAL TRANSLATION, GOVERNANCE AND MONITORING DESIGN

EXTERNAL VALIDATION — NOT ESTABLISHED

CLINICAL EFFECTIVENESS — NOT ESTABLISHED

PRODUCTION CLINICAL USE — NOT AUTHORIZED

10. REVIEW AND SIGN-OFF

Reviewers confirm that the validation evidence, limitations, residual risks, and governance disposition documented in this report have been reviewed.

Sign-off does not independently constitute authorization for production clinical deployment.

Role	Name	Decision	Signature	Date
Project & Model Owner				
Clinical Validation Lead				
AI Governance Reviewer				
Independent Reviewer				
Approval Authority				

DOCUMENT CONTROL

Field	Value
Document	Clinical AI Model Validation Report
Project	Diabetes Readmission Clinical AI
Model ID	DIABETES_READMISSION_XGB_D13_V1
Version	1.0.0
Validation Stage	Internal Locked-Test Validation
Frozen Threshold	0.12
Governance Disposition	Conditional Pass

External Validation	Not Established
Clinical Effectiveness	Not Established
Production Deployment	Not Authorized